CHAPTER 9

Resonant Converters

9.1 INTRODUCTION

Imperfect switching is a major contributor to power loss in converters, as discussed in Chap. 6. Switching devices absorb power when they turn on or off if they go through a transition when both voltage and current are nonzero. As the switching frequency increases, these transitions occur more often and the average power loss in the device increases. High switching frequencies are otherwise desirable because of the reduced size of filter components and transformers, which reduces the size and weight of the converter.

In resonant switching circuits, switching takes place when voltage and/or current is zero, thus avoiding simultaneous transitions of voltage and current and thereby eliminating switching losses. This type of switching is called *soft* switching, as opposed to *hard* switching in circuits such as the buck converter. Resonant converters include resonant switch converters, load resonant converters, and resonant dc link converters. This chapter introduces the basic concept of the resonant converter and gives a few examples.

9.2 A RESONANT SWITCH CONVERTER: ZERO-CURRENT SWITCHING

Basic Operation

One method for taking advantage of the oscillations caused by an LC circuit for reducing the switching losses in a dc-dc converter is shown in the circuit of Fig. 9-1*a*. This circuit is similar to the buck converter described in Chap. 6. The current in the output inductor L_o is assumed to be ripple-free and equal to the load

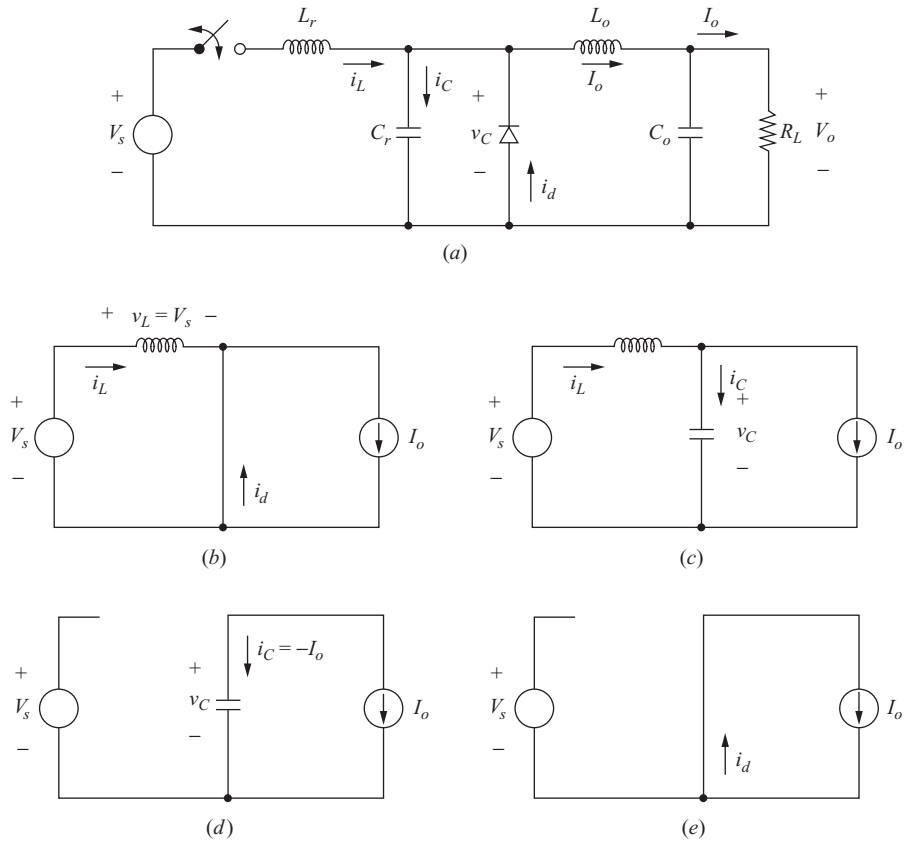
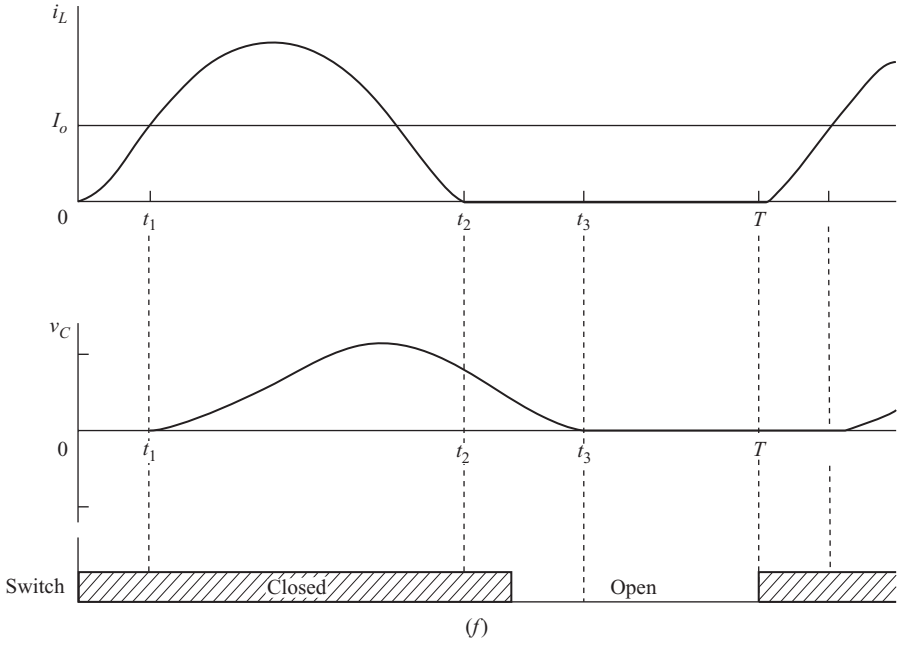
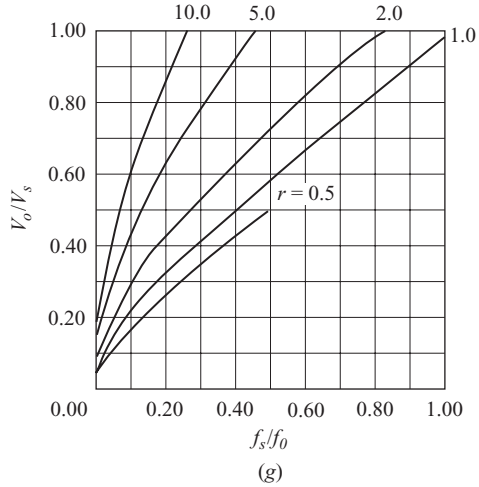


Figure 9-1 (*a*) A resonant converter with zero-current switching; (*b*) Switch closed and diode on ($0 < t < t_1$); (*c*) Switch closed and diode off ($t_1 < t < t_2$); (*d*) Switch open and diode off ($t_2 < t < t_3$); (*e*) Switch open and diode on ($t_3 < t < T$); (*f*) Waveforms; (*g*) Normalized output vs. switching frequency with $r = R_L/Z_0$ as a parameter. © 1992 IEEE, B.K. Bose, Modern Power Electronics: Evolution, Technology, and Applications. Reprinted with permission.

current I_o. When the switch is open, the diode is forward-biased to carry the output inductor current, and the voltage across C_r is zero. When the switch closes, the diode initially remains forward-biased to carry I_o, and the voltage across L_r is the same as the source voltage V_s (Fig. 9-1*b*). The current in L_r increases linearly, and the diode remains forward-biased while i_L is less than I_o. When i_L reaches I_o, the diode turns off, and the equivalent circuit is that of Fig. 9-1*c*. If I_o is a constant, the load appears as a current source, and the underdamped LC circuit oscillates. Consequently, i_L returns to zero and remains there, assuming the switch is unidirectional. The switch is turned off after the current reaches zero, resulting in zero-current switching and no switching power loss.



(*f*)



(*g*)

Figure 9-1 (*continued***)**

After the current in the switch reaches zero, the positive capacitor voltage keeps the diode reverse-biased, so load current I_o flows through C_r, with $i_c = -I_o$ (Fig. 9-1*d*). If I_o is constant, the capacitor voltage decreases linearly. When the capacitor voltage reaches zero, the diode becomes forward-biased to carry I_o (Fig. 9-1*e*). The circuit is then back at the starting point. The analysis for each time interval is given next.

Analysis for $0 \le t \le t_1$ The switch is closed at $t = 0$, the diode is on, and the voltage across L_r is V_s (Fig. 9-1*b*). The current in L_r is initially zero and is expressed as

$$i_L(t) = \frac{1}{L_r}\int_0^t V_s\, d\lambda = \frac{V_s}{L_r}t \tag{9-1}$$

At $t = t_1$, i_L reaches I_o, and the diode turns off. Solving for t_1,

$$i_L(t_1) = I_o = \frac{V_s}{L_r}t_1 \tag{9-2}$$

or

$$\boxed{t_1 = \frac{I_o L_r}{V_s}} \tag{9-3}$$

Capacitor voltage is zero in this interval.

Analysis for $t_1 \le t \le t_2$ (Fig. 9-1*c*) When the diode turns off at $t = t_1$, the circuit is equivalent to that in Fig. 9-1*c*. In the circuit of Fig. 9-1*c*, these equations apply:

$$v_C(t) = V_s - L_r\frac{di_L(t)}{dt} \tag{9-4}$$

$$i_C(t) = i_L(t) - I_o \tag{9-5}$$

Differentiating Eq. (9-4) and using the voltage-current relationship for the capacitor,

$$\frac{dv_C(t)}{dt} = -L_r\frac{d^2i_L(t)}{dt^2} = \frac{i_C(t)}{C_r} \tag{9-6}$$

Substituting for i_C using Eq. (9-5),

$$L_r\frac{d^2i_L(t)}{dt^2} = \frac{I_o - i_L(t)}{C_r} \tag{9-7}$$

or

$$\frac{d^2i_L(t)}{dt^2} + \frac{i_L(t)}{L_rC_r} = \frac{I_o}{L_rC_r} \tag{9-8}$$

The solution to the preceding equation with the initial condition $i_L(t_1) = I_o$ is

$$i_L(t) = I_o + \frac{V_s}{Z_o}\sin\omega_0(t - t_1) \tag{9-9}$$

where Z_0 is the characteristic impedance

$$Z_0 = \sqrt{\frac{L_r}{C_r}} \tag{9-10}$$

and ω_0 is the frequency of oscillation

$$\omega_0 = \frac{1}{\sqrt{L_r C_r}} \tag{9-11}$$

Equation (9-9) is valid until i_L reaches zero at $t = t_2$. Solving for the time interval $t_2 - t_1$ when the oscillation occurs,

$$t_2 - t_1 = \frac{1}{\omega_0} \sin^{-1}\left(\frac{-I_o Z_0}{V_s}\right) \tag{9-12}$$

which can be expressed as

$$\boxed{t_2 - t_1 = \frac{1}{\omega_0}\left[\sin^{-1}\left(\frac{I_o Z_0}{V_s}\right) + \pi\right]} \tag{9-13}$$

Solving for capacitor voltage by substituting i_L from Eq. (9-9) into Eq. (9-4) gives

$$v_C(t) = V_s\{1 - \cos[\omega_0(t - t_1)]\} \tag{9-14}$$

which is also valid until $t = t_2$. Maximum capacitor voltage is therefore $2V_s$.

Analysis for $t_2 \le t \le t_3$ After the inductor current reaches zero at t_2, switch current is zero and it can be opened without power loss. The equivalent circuit is shown in Fig. 9-1*d*. The diode is off because $v_C > 0$. Capacitor current is $-I_o$, resulting in a linearly decreasing capacitor voltage expressed as

$$v_C(t) = \frac{1}{C_r}\int_{t_2}^{t} -I_o\, d\lambda + v_C(t_2) = \frac{I_o}{C_r}(t_2 - t) + v_C(t_2) \tag{9-15}$$

Equation (9-15) is valid until the capacitor voltage reaches zero and the diode turns on. Letting the time at which the capacitor voltage reaches zero be t_3, Eq. (9-15) gives an expression for the time interval $t_3 - t_2$:

$$\boxed{t_3 - t_2 = \frac{C_r v_C(t_2)}{I_o} = \frac{C_r V_s\{1 - \cos[\omega_0(t_2 - t_1)]\}}{I_o}} \tag{9-16}$$

where $v_C(t_2)$ is obtained from Eq. (9-14).

Analysis for $t_3 \le t \le T$ In this time interval, i_L is zero, the switch is open, the diode is on to carry I_o, and $v_C = 0$ (Fig. 9-1*e*). The duration of this interval is the difference between the switching period T and the other time intervals, which are determined from other circuit parameters.

Output Voltage

Output voltage can be determined from energy balance. Energy supplied by the source is equal to energy absorbed by the load during a switching period. Energy supplied by the source in one period is

$$W_s = \int_0^T p_s(t)\,d(t) = V_s \int_0^T i_L(t)\,dt \tag{9-17}$$

Energy absorbed by the load is

$$W_o = \int_0^T p_o(t)\,dt = V_o I_o T = \frac{V_o I_o}{f_s} \tag{9-18}$$

where f_s is the switching frequency. From Eqs. (9-1) and (9-9),

$$\int_0^T i_L(t)\,dt = \int_0^{t_1} \frac{V_s t}{L_r}\,dt + \int_{t_1}^{t_2} \left\{ I_o + \frac{V_s}{Z_0} \sin[\omega_0(t - t_1)] \right\} dt \tag{9-19}$$

Using $W_s = W_o$ and solving for V_o using Eqs. (9-17) to (9-19),

$$V_o = V_s f_s \left(\frac{t_1}{2} + (t_2 - t_1) + \frac{V_s C_r}{I_o} \{1 - \cos[\omega_0(t_2 - t_1)]\} \right) \tag{9-20}$$

Using Eq. (9-16), output voltage can be expressed in terms of the time intervals for each circuit condition:

$$\boxed{V_o = V_s f_s \left[\frac{t_1}{2} + (t_2 - t_1) + (t_3 - t_2) \right]} \tag{9-21}$$

where the time intervals are determined from Eqs. (9-3), (9-13), and (9-16).

Equation (9-21) shows that the output voltage is a function of the switching frequency. Increasing f_s increases V_o. The switching period must be greater than t_3, and output voltage is less than input voltage, as is the case for the buck converter of Chap. 6. Note that the time intervals are a function of output current I_o, so output voltage for this circuit is load-dependent. When the load is changed, the switching frequency must be adjusted to maintain a constant output voltage. Figure 9-1*g* shows the relationship between output voltage and switching frequency. The quantity $r = R_L/Z_0$ is used as a parameter where R_L is the load resistance and Z_0 is defined in Eq. (9-10).

A diode placed in antiparallel with the switch in Fig. 9-1*a* creates a resonant switch converter which includes negative inductor current. For that circuit, V_o/V_s is nearly a linear function of switching frequency independent of load (that is, $V_o/V_s = f_s/f_0$).

The resonant switch converter with zero-current switching has theoretically zero switching losses. However, junction capacitance in switching devices stores energy which is dissipated in the device, resulting in small losses.

Note that output voltage is the average of the capacitor voltage v_c, yielding an alternate method of deriving Eq. (9-21).

EXAMPLE 9-1

Resonant Switch DC-DC Converter: Zero-Current Switching

In the circuit of Fig. 9-1*a*,

$$V_s = 12 \text{ V}$$
$$C_r = 0.1 \ \mu\text{F}$$
$$L_r = 10 \ \mu\text{H}$$
$$I_o = 1 \text{ A}$$
$$f_s = 100 \text{ kHz}$$

(*a*) Determine the output voltage of the converter. (*b*) Determine the peak current in L_r and the peak voltage across C_r. (*c*) What is the required switching frequency to produce an output voltage of 6 V for the same load current? (*d*) Determine the maximum switching frequency. (*e*) If the load resistance is changed to 20 Ω, determine the switching frequency required to produce an output voltage of 8 V.

■ Solution

(*a*) Using the given circuit parameters,

$$\omega_0 = \frac{1}{\sqrt{L_r C_r}} = \frac{1}{\sqrt{10(10)^{-6}(0.1)(10)^{-6}}} = 10^6 \text{ rad/s}$$

$$Z_0 = \sqrt{\frac{L_r}{C_r}} = \sqrt{\frac{10(10)^{-6}}{0.1(10)^{-6}}} = 10 \ \Omega$$

Output voltage is determined from Eq. (9-21). The time t_1 is determined from Eq. (9-3):

$$t_1 = \frac{I_o L_r}{V_s} = \frac{(1)(10)(10)^{-6}}{12} = 0.833 \ \mu\text{s}$$

From Eq. (9-13),

$$t_2 - t_1 = \frac{1}{\omega_0}\left[\sin^{-1}\left(\frac{I_o Z_0}{V_s}\right) + \pi\right] = \frac{1}{10^6}\left[\sin^{-1}\frac{(1)(10)}{12} + \pi\right] = 4.13 \ \mu\text{s}$$

From Eq. (9-16),

$$t_3 - t_2 = \frac{C_r V_s}{I_o}\{1 - \cos[\omega_0(t_2 - t_1)]\}$$

$$= \frac{(0.1)(10)^{-6}(12)}{1}\{1 - \cos[10^6(4.13)(10)^{-6}]\} = 1.86 \ \mu\text{s}$$

Output voltage from Eq. (9-21) is then

$$V_o = V_s f_s \left[\frac{t_1}{2} + (t_2 - t_1) + (t_3 - t_2) \right]$$

$$= (12)(100)(10^5) \left(\frac{0.833}{2} + 4.13 + 1.86 \right)(10^{-6}) = 7.69 \text{ V}$$

(*b*) Peak current in L_r is determined from Eq. (9-9).

$$I_{L,\text{peak}} = I_o + \frac{V_s}{Z_0} = 1 + \frac{12}{10} = 2.2 \text{ A}$$

Peak voltage across C_r is determined from Eq. (9-14):

$$V_{C,\text{peak}} = 2V_s = 2(12) = 24 \text{ V}$$

(*c*) Since output voltage is proportional to frequency [Eq. (9-21)] if I_o remains unchanged, the required switching frequency for a 6-V output is

$$f_s = 100 \text{ kHz} \left(\frac{6 \text{ V}}{7.69 \text{ V}} \right) = 78 \text{ kHz}$$

(*d*) Maximum switching frequency for this circuit occurs when the interval $T - t_3$ is zero. Time $t_3 = t_1 + (t_2 - t_1) + (t_3 - t_2) = (0.833 + 4.13 + 1.86)\ \mu\text{s} = 6.82\ \mu\text{s}$, resulting in

$$f_{s,\max} = \frac{1}{T_{\min}} = \frac{1}{t_3} = \frac{1}{(6.82)(10^{-6})} = 146 \text{ kHz}$$

(*e*) The graph of Fig. 9-1*g* can be used to estimate the required switching frequency to obtain an output of 8 V with the load at 20 Ω. With $V_o/V_s = 8/12 = 0.67$, the curve for the parameter $r = R_L/Z_0 = 20/10 = 2$ gives $f_s/f_0 \approx 0.45$. The switching frequency is $f_s = 0.45 f_0 = 0.45(\omega_0/2\pi) = 0.45(10)^6/2\pi = 71.7$ kHz. The method used in part (*a*) of this problem can be used to verify the results. Note that I_o is now $V_o/R_L = 8/20 = 0.4$ A.